

# DESIGN EXPLANATION

## Adaptive Power Distribution System

*12 V DC Input • Regulated 5 V & 3.3 V Outputs*

### 1. Design Objective

The objective of the Adaptive Power Distribution System is to accept a 12 V DC input and distribute regulated low-voltage power to multiple loads. The design provides a regulated 5 V rail for two loads and a regulated 3.3 V rail for one load, while incorporating input protection, reverse-polarity protection, current protection, power indication and load/status indication.

The design was developed as a complete electronics system: the circuit was simulated in Multisim, the schematic and PCB were developed in KiCad, and the component ratings and power requirements were considered during the design process.

### 2. Overall Circuit Architecture

The system is divided into the following functional blocks:

| Block                            | Function                                                                     |
|----------------------------------|------------------------------------------------------------------------------|
| 12 V DC Input                    | Accepts the external 12 V supply through the input connector.                |
| Input Protection                 | Provides input overcurrent protection and reverse-polarity protection.       |
| Input Filtering / Energy Storage | Uses bulk and high-frequency capacitors on the protected input rail.         |
| 5 V Buck Converter               | LM2596S-5.0 generates the regulated 5 V supply.                              |
| 3.3 V Buck Converter             | LM2596S-3.3 generates the regulated 3.3 V supply.                            |
| Output Protection                | Polyfuses provide resettable overcurrent protection on the output branches.  |
| Power / Load Indicators          | LED indicator branches provide visual indication for the input/output rails. |
| Output Connectors                | Distribute the regulated rails to the specified loads.                       |

Power flows from the 12 V input through the input protection stage and then to both buck-converter stages. The converters independently generate the required 5 V and 3.3 V rails. Each output is then distributed through protected branches to the corresponding load connectors.

### 3. Input Protection and Power Entry

#### 3.1 Input overcurrent protection

A 3 A fuse is placed in series with the 12 V input. Its purpose is to interrupt the input supply in the event of an abnormal overcurrent condition and to protect the downstream circuitry.

#### 3.2 Reverse-polarity protection

A 1N5822 Schottky diode is used in series with the input path for reverse-polarity protection. Under normal polarity the diode permits the supply to reach the protected 12 V rail; if the supply polarity is reversed, the diode blocks the reverse connection.

#### 3.3 Input filtering and energy storage

The protected input rail includes a 470  $\mu\text{F}$  bulk capacitor and a 100 nF bypass capacitor. The bulk capacitor provides local energy storage and helps reduce input-voltage variation, while the 100 nF capacitor provides high-frequency bypassing.

### 4. Voltage Regulation Approach

Two independent buck-converter stages are used rather than a single regulator because the system requires two different regulated output voltages.

| Stage       | Regulator   | Output | Required load         |
|-------------|-------------|--------|-----------------------|
| 5 V stage   | LM2596S-5.0 | 5.0 V  | 0.5 A + 1.0 A = 1.5 A |
| 3.3 V stage | LM2596S-3.3 | 3.3 V  | 0.3 A                 |

The buck topology is appropriate because both required output voltages are lower than the 12 V input. A switching regulator is also better suited to the specified power level than a linear regulator because it can reduce conversion power dissipation.

### 5. Buck-Converter Power Stage

Each LM2596 stage uses the required switching-converter elements: the regulator, a 68  $\mu\text{H}$  inductor, a 1N5822 Schottky diode and a 330  $\mu\text{F}$  output capacitor. The output capacitor supports transient load demand and helps stabilize the regulated rail.

The switching nodes were treated as high-frequency nodes during PCB routing. Their traces were kept short to reduce unnecessary loop area. The higher-current 12 V, 5 V and 3.3 V distribution paths were assigned wider PCB traces, while small-signal connections use narrower traces.

### 6. Output Power Distribution

The 5 V rail supplies two specified loads: Load A at 500 mA and Load C at 1 A. Their combined nominal current is therefore 1.5 A. The 3.3 V rail supplies Load B at 300 mA.

| Output | Loads        | Total current | Nominal output power |
|--------|--------------|---------------|----------------------|
| 5 V    | 500 mA + 1 A | 1.50 A        | 7.50 W               |

|       |                     |        |        |
|-------|---------------------|--------|--------|
| 3.3 V | 300 mA              | 0.30 A | 0.99 W |
| Total | All specified loads | —      | 8.49 W |

## 7. Output Current Protection

Resettable polyfuses are placed before the output connectors. This creates separately protected output branches rather than relying only on a common input fuse. The selected branches are intended for the specified 5 V/0.5 A, 5 V/1 A and 3.3 V/0.3 A loads.

The use of separate protected branches improves fault isolation: an abnormal condition on one output branch can be limited without requiring the entire distribution system to be disconnected. The exact hold and trip characteristics depend on the final polyfuse part selected and its operating temperature.

## 8. Power and Load Indication

LED indicator branches are included to provide visual status information for the available rails. Series resistors limit LED current. The indicators are intentionally treated as low-power branches and therefore use narrow signal traces compared with the main power rails.

## 9. Schematic Design Decisions

| Decision                      | Reason                                                                                    |
|-------------------------------|-------------------------------------------------------------------------------------------|
| Two independent buck stages   | Required 5 V and 3.3 V outputs are independently regulated.                               |
| Schottky input diode          | Provides simple reverse-polarity protection with relatively low forward drop.             |
| 3 A input fuse                | Provides input overcurrent protection with margin over the expected normal input current. |
| 470 $\mu$ F input capacitor   | Provides bulk energy storage and input smoothing.                                         |
| 100 nF bypass capacitors      | Provides high-frequency local bypassing.                                                  |
| 330 $\mu$ F output capacitors | Supports output transients and reduces output ripple.                                     |
| Output polyfuses              | Provides resettable branch-level current protection.                                      |
| Separate output connectors    | Allows independent distribution of the required rails to loads.                           |

## 10. PCB Architecture

The PCB was organized so that the power-entry and protection section is separated from the two converter stages and the output connectors. The inductors, Schottky diodes and output capacitors are kept close to their respective LM2596 stages to keep the switching paths compact.

High-current power nets were routed with wider traces. The 12 V main path uses approximately 1.5 mm, the 5 V main path approximately 1.0–1.5 mm depending on the branch, and the 3.3 V main path approximately 1.0 mm. Small signal paths use approximately 0.2–0.25 mm. A copper ground plane is used to provide a low-impedance return path and simplify ground distribution.

The completed PCB concept also includes output polyfuses, output connectors and status indicators, providing a compact power-distribution arrangement rather than a collection of independent regulator circuits.

## 11. Simulation Approach

The circuit was simulated in Multisim under the specified resistive load conditions. The load resistances were calculated from  $R = V/I$ : 10  $\Omega$  for 5 V/500 mA, 5  $\Omega$  for 5 V/1 A, and 11  $\Omega$  for 3.3 V/300 mA.

Voltage probes and current measurements were used to verify the input and regulated output behavior. An oscilloscope was also used to observe the output waveform and settling behavior. The simulation confirmed that the converter stages reach the intended regulated output levels under the modeled load conditions.

During simulation, a long real-time observation was initially noticed at the output measurement display. The oscilloscope measurement showed the electrical response occurring on a much shorter simulation-time scale, so the long timer observation was treated as a simulation/display timing issue rather than as a physical one-minute-plus regulator startup requirement.

## 12. Design Verification

| Item                    | Target / requirement | Design implementation                |
|-------------------------|----------------------|--------------------------------------|
| Input voltage           | 12 V DC              | 12 V DC input connector              |
| 5 V output              | 5 V regulated        | LM2596S-5.0                          |
| 3.3 V output            | 3.3 V regulated      | LM2596S-3.3                          |
| 5 V load capacity       | 1.5 A total          | Dedicated 5 V buck stage             |
| 3.3 V load capacity     | 0.3 A                | Dedicated 3.3 V buck stage           |
| Input protection        | Required             | 3 A fuse + 1N5822                    |
| Current protection      | Required             | Output polyfuses                     |
| Power indication        | Required             | LED indicators                       |
| Safe power distribution | Required             | Protected individual output branches |

## 13. Limitations

- Reverse-polarity protection using a series Schottky diode introduces a forward-voltage drop and associated power loss.
- Exact converter efficiency and thermal performance depend on the selected LM2596 implementation, inductor, diode, capacitors, PCB copper area and operating temperature.

- Polyfuse behavior is temperature- and time-dependent; the final part's hold/trip curves should be checked against the intended loads.
- The Multisim model does not replace hardware validation. Real measurements should be performed for ripple, efficiency, temperature rise, startup behavior and protection response.
- The design is intended for the specified low-voltage loads; higher loads should not be connected without rechecking regulator current capability, thermal performance, traces, connectors and protection ratings.

## 14. Possible Improvements

- Replace the series reverse-polarity diode with an ideal-diode MOSFET arrangement to reduce input voltage drop and conduction loss.
- Use regulators with integrated thermal/current protection and optimize the power-stage components using the exact regulator datasheet recommendations.
- Add dedicated input and output transient protection such as TVS protection if the real operating environment contains significant surges or cable transients.
- Add test points for 12 V protected, 5 V, 3.3 V and GND to simplify production testing and debugging.
- Add mounting holes and mechanical keep-outs for enclosure integration.
- Validate the finished PCB experimentally using electronic loads and a thermal camera or thermocouple, then update the efficiency and thermal calculations with measured data.
- Consider separate status indicators for each protected output branch if more detailed load-status information is required.

## 15. Conclusion

The final design implements a compact 12 V DC power-distribution system with independently regulated 5 V and 3.3 V outputs. The architecture combines buck conversion, input protection, reverse-polarity protection, branch-level resettable current protection, filtering, indication and a dedicated PCB layout. The calculated full-load output requirement is 8.49 W, consisting of 7.50 W on the 5 V rail and 0.99 W on the 3.3 V rail. The completed schematic, Multisim simulation and PCB were developed around these requirements.